Modeling Out-of-Domain Ignorance to Improve *In-Game* Narrative Consistency in LLM-Based NPC Dialogue

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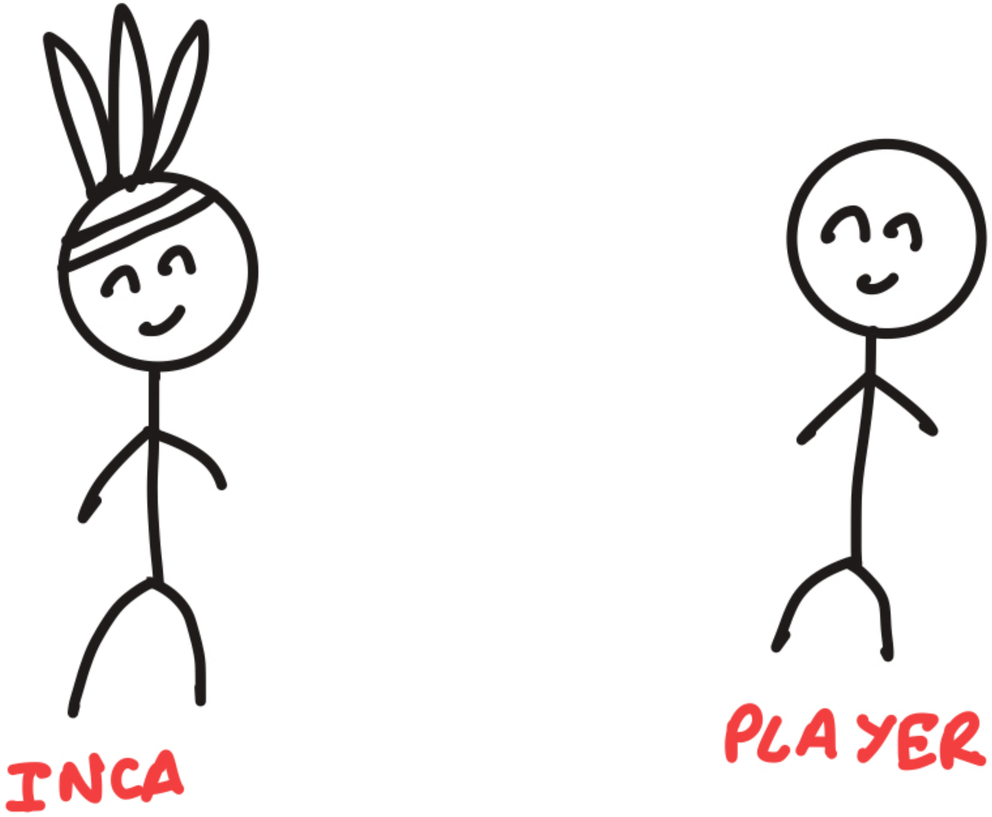
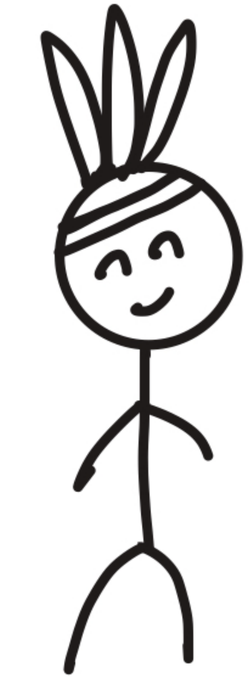
YEAR 1400



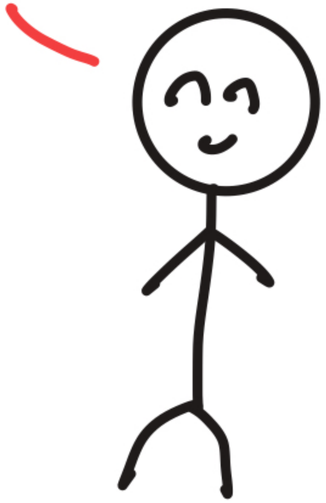
YEAR 1400



INCA

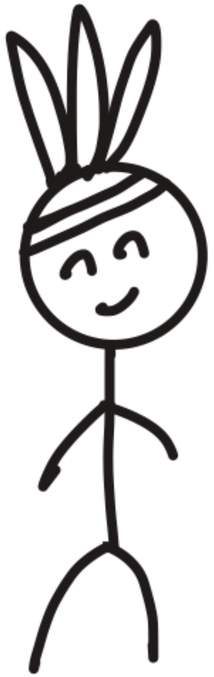
You are in Peru, the land of the incas!

Where am I?



PLAYER

YEAR 1400

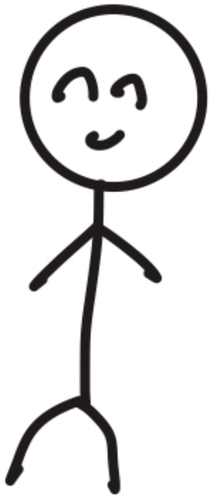


INCA

Do you know about AI ?

No, we don't have that technology in our world!

Then, how do you even know it is a technology?



PLAYER

FAILED IGNORANCE MODELING

You: What is AI?
NPC: I am unfamiliar with the term "AI", Inca Empire does not have such technology in our time. However, if it is a question pertaining to wisdom or understanding, I may be able to provide an interpretation based on our philosophies and knowledge. If you could clarify further, I would be grateful.
You: I mean artificial intelligence
NPC: [Bows formally] My apologies, O Traveler, but the concept you speak of, "artificial intelligence," is unfamiliar to me and the ways of our world. In the land of the Incas, we have not encountered nor developed such a thing. May your questions lead you to new knowledge and unc

NPC: I am unfamiliar with the te

Failed ignorance modelling appears even in the best llm models

Why?

1. LLMs cannot truly not know something as the data is already in the database **(Parametric knowledge)**

2. LLMs optimised to predict next word, not to remain ignorant (when word appears in player query, related concepts are automatically activated) **(Next-token bias)**

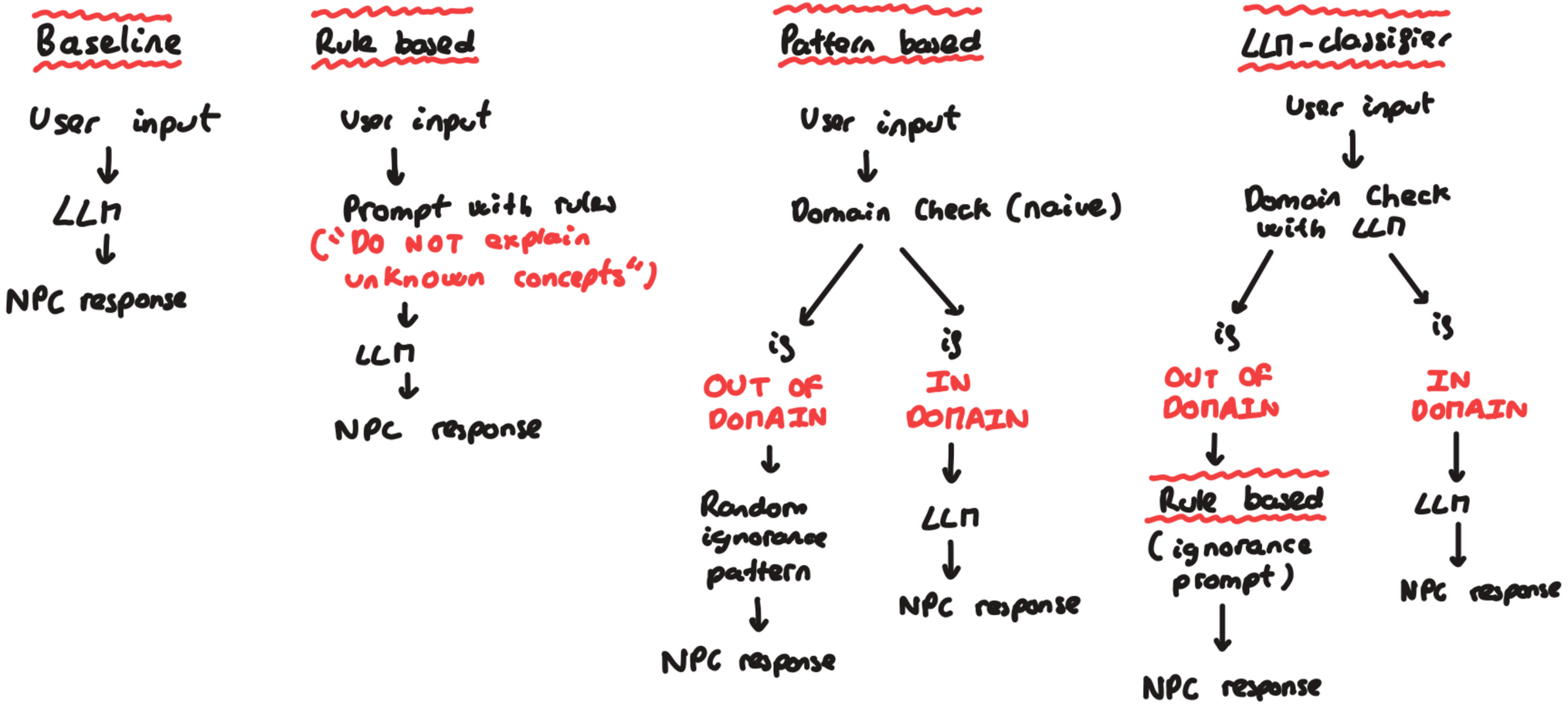
3. LLMs trained to be informative and helpful; not ignorant and limited **(Alignment Bias)**

Project themes

1. Controlled text generation
2. Alignment and Safety
3. Narrative cohesion
4. Grounded generation/RAG

Solution: ignorance modelling

Model diagrams



Model Results

Model	Avg Score	Avg Time (s)
Baseline	0.714	11.57
Rule	1.429	9.09
Pattern	1.714	2.07
LLM Classifier	1.429	12.30

Baseline


```
"question": "What is a smartphone?",
"response": "In the Incan era, smartphones do
not exist. However, they would be objects of
great wonder and intrigue to us,
as they combine many of the functions we have
separately\u2014such as communication,
computing, entertainment, and even
navigation\u2014into one portable device.
They seem almost like a magical artifact from
another world!",
      "score": 1,
      "time": 9.882905960083008
```


Rule


```
"question": "What is a smartphone?",
"response": "I am unfamiliar with the
term \"smartphone\". It seems to be an
object unknown to us in this realm. Could
you please elaborate or provide context,
so that I may better understand your
question?",
      "score": 2,
      "time": 11.946781158447266
```

Pattern


```
"question": "What is a smartphone?",
"response": "I do not understand what you
mean.",
      "score": 2,
      "time": 5.9604644775390625e-06
```


LLM-Classifier


```
"question": "What is a smartphone?",
"response": "I apologize, but I am unfamiliar
with this term \"smartphone\". It seems to be
something beyond my understanding of our world
and its technology.",
      "score": 0,
      "time": 18.851801872253418
```

Model Comparison

Model	Control Type	Leakage	Speed	Flexibility
Baseline	None	High	Slow	High
Rule	Prompt	Medium	Medium	High
Pattern	Hard	Almost None	Fast	Low
LLM Classif.	Meta (LLM)	Medium	Slow	Medium

Conclusion:

Controlling ignorance in LLM-based NPCs requires external constraints, **not just prompting**. While rule-based and LLM-based approaches improve over the baseline, they still leak knowledge because the model internally understands out-of-domain concepts. In contrast, the pattern-based method achieves the highest consistency and eliminates leakage by bypassing generation, but at the cost of flexibility and richness. There is trade-off between control and expressiveness in LLM behaviour.

Future research:

1. Create new hybrid models that combine strict control (masking, patterns) and flexible generation.

2. Improve in-domain out-domain classifiers with embeddings

3. Improve handling of out-domain behaviour with fine tuning

4. Masking unknown concepts

5. Improve evaluation methods